SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE, UNLIMITED POTENTIAL

Department of Computer Science & Engineering
Problem Solving Using C-25BECPHY104
AY-2025-26 Sem-I
Question Bank

MODULE-1				
Sl.No	**Question**	**RBT Level**	**COs**	**Marks**
1.	Explain the basic structure of a C program.	L2	CO1	10
2.	What is a variable? Explain different rules to define a variable name with examples.	L2	CO1	5
3.	List different characteristics of C programming	L2	CO1	5
4.	What are identifiers? State the rules for naming them.	L3	CO1	7
5.	Differentiate between variables and identifier.	L3	CO1	5
6.	Explain the use of `scanf()` with an example.	L2	CO1	5
7.	Write a short note on data types in C.	L2	CO1	10
8.	What are format specifiers? List any four with their meanings.	L3	CO1	5
9.	What are operators? Classify the types of operators in C.	L2	CO1	10
10.	Explain the concept of type casting with an example.	L3	CO1	5
11.	Describe the steps involved in compiling and executing a C program.	L2	CO1	5
12.	Explain in detail the different types of data types in C with examples.	L3	CO1	10
13.	Write a C program to read two numbers and display their sum, difference, product, and quotient.	L2	CO1	10
14.	Discuss the importance of header files in C programming.	L2	CO1	5
15.	Explain the difference between global and local variables with examples.	L2	CO1	5
16.	Describe different types of constants in C with suitable examples.	L2	CO1	5
17.	Write and explain a C program to calculate the area of a circle.	L2	CO1	10
18.	Explain the structure of a C program using an example and label each part.	L2	CO1	10
20.	Describe the steps involved in compiling and executing a C program.	L2	CO1	10

22.	Design an algorithm and flowchart and then write a program to find the sum of two numbers.	L2	CO1	10
23.	Design an algorithm and flowchart and then write a program to find the average of three numbers entered by the user.	L2	CO1	10
24.	Design an algorithm and flowchart and then write a program to Check whether a person is eligible to vote.	L2	CO1	10
25.	Design an algorithm and flowchart and then write a program to find the area and perimeter of a rectangle.	L2	CO1	10
26.	Design an algorithm and flowchart and then write a program to swap two numbers using a temporary variable.	L2	CO1	10
27.	Design an algorithm and flowchart and then write a program to calculate simple interest.	L2	CO1	10
28.	Design an algorithm and flowchart and then write a program to print your name, roll number, and college name.	L2	CO1	10
29.	Design an algorithm and flowchart and then write a program to find the remainder and quotient of two integers.	L2	CO1	10
30.	Define an Algorithm. List out different advantages and features of Algorithm.	L2	CO1	10
31.	What is the procedure to design an algorithm in C.	L1	CO1	05
32.	Define Flowchart. Explain different symbols used in flowchart.	L2	CO1	10